



NTNU – Trondheim
Norwegian University of
Science and Technology

TTT4120 Digital Signal Processing Fall 2025

Lecture: Discrete Time Systems in Frequency Domain

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Lecture in course book*

- Proakis, Manolakis Digital Signal Processing, 4th Ed.
 - 4.2.1 Fourier series for discrete-time periodic signals
 - 4.2.3 Fourier transform of discrete-time aperiodic signals
 - 4.3 Frequency-domain and time-domain signal properties
 - 5.1.1 Response to complex exponential and sinusoidal...
 - 5.1.4 Response to aperiodic input signals
 - 5.4.1 Ideal filter characteristics

*Level of detail is defined by lectures and problem sets

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Contents and learning outcomes

- Discrete-time Fourier series (DTFS) for periodic signals
- Discrete-time Fourier transform (DTFT) for aperiodic signals
- Properties of the Fourier transform
- LTI systems in frequency domain – the frequency response function $H(\omega)$

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Frequency analysis of DT signals

- The impulse response of a linear time-invariant system $h[n]$ allows us to compute the response to an arbitrary input $x[n]$

$$\begin{array}{c}
 x[n] \longrightarrow \boxed{h[n]} \longrightarrow y[n] = h[n] * x[n] \\
 \hspace{15em} = \sum_k h[k]x[n-k]
 \end{array}$$

- Convolution sum is based on the fact that **any input sequence** can be **decomposed as a linear combination** of **scaled and delayed unit impulse sequences**, $x[n] = \sum_k x[k]\delta[n-k]$
- We can choose to represent the signal using a linear combination of some **other basis signals**

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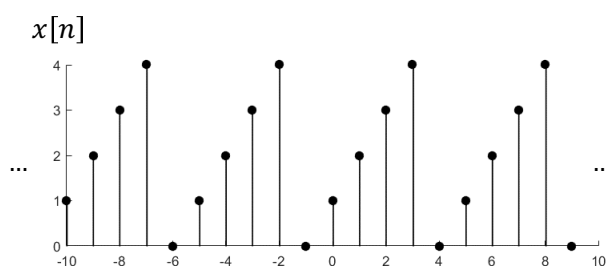
Frequency analysis of DT signals...

- Most signals of practical interest can be decomposed into a sum of complex exponentials
- Using such a combination, a signal is said to be represented in **frequency domain**
 - periodic signals $\Rightarrow$ discrete-time Fourier series (DTFS)
 - finite-energy signals $\Rightarrow$ DT Fourier transform (DTFT)
- We shall see that this decomposition is very important in the analysis of linear time-invariant systems
 - response to a sinusoidal input signal is a sinusoid with the **same frequency** but **different amplitude and phase**
 - linear combination of sinusoids at input produces a similar linear combination of sinusoids at output

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Discrete-time Fourier series (DTFS)



- Discrete-time signal $x[n]$ periodic with period N

$$x[n + N] = x[n], \forall n$$

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Discrete-time Fourier series (DTFS)

- Fourier series representation for $x[n]$ consists of a weighted sum of N harmonically related exponentials $e^{j2\pi k n/N}$

$$x[n] = \sum_{k=0}^{N-1} c_k e^{j2\pi k n/N} \text{ (synthesis equation)}$$

- Fourier coefficients c_k provide frequency-domain information of $x[n]$ and are given by

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j2\pi k n/N} \text{ (analysis equation)}$$

- Spectrum of **periodic** sequence is **periodic**

$$c_{k+N} = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j2\pi(k+N)n/N} = c_k$$

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Discrete-time Fourier series (DTFS)...

- Only need to concentrate on a single period in frequency

$$0 \leq \omega_k < 2\pi \quad \text{or} \quad -\pi \leq \omega_k < \pi$$

$$\text{with } \omega_k = 2\pi k/N$$

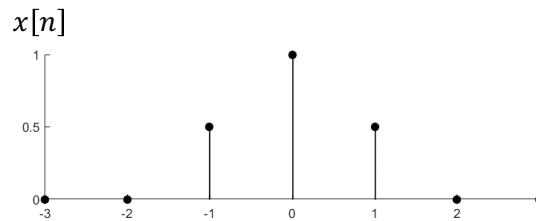
- Periodic signal in time-domain $\Rightarrow$ discrete spectrum
- Example: $x_1[n] = \cos \pi^2 n$

$$x_2[n] = \cos \frac{\pi n}{4}$$

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Discrete-time Fourier transform (DTFT)



- Discrete-time signal $x[n]$ is *aperiodic* but has *finite energy*

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Discrete-time Fourier transform (DTFT)

- Discrete-time Fourier transform (DTFT) of $x[n]$:

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \text{ (analysis equation)}$$

- Represents the frequency content of $x[n]$ and is 2π -periodic

$$X(\omega + 2\pi k) = X(\omega)$$

- Frequency range for any discrete-time signal $x[n]$ is limited to $(-\pi, \pi)$ or $(0, 2\pi)$
- We may obtain $x[n]$ from $X(\omega)$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega \text{ (synthesis equation)}$$

- Notation: $x[n] \xleftrightarrow{\mathcal{F}} X(\omega)$

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Discrete-time Fourier transform (DTFT)...

- Examples: $x_1[n] = \delta[n] \xleftrightarrow{\mathcal{F}} X_1(\omega) = ?$

$$x_2[n] = ? \xleftrightarrow{\mathcal{F}} X_2(\omega) = \delta(\omega)$$

$$x_3[n] = a^n u[n] \xleftrightarrow{\mathcal{F}} X_3(\omega) = ?$$

$$x_4[n] = ? \xleftrightarrow{\mathcal{F}} X_4(\omega) = \begin{cases} 1, & |\omega| \leq \omega_c < \pi \\ 0, & \text{otherwise} \end{cases}$$

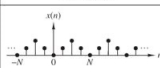
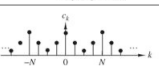
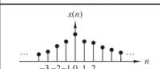
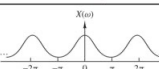
$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$$

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Summary DTFS and DTFT

Discrete-time signals	
Time-domain	Frequency-domain
 $c_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$	 $x[n] = \sum_{k=-\infty}^{\infty} c_k e^{j2\pi kn/N}$
Discrete and periodic	Discrete and periodic
 $X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$	 $x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$
Discrete and aperiodic	Continuous and periodic

- Discrete-time signals have periodic spectra
- Periodic signals $\Rightarrow$ discrete spectra $\omega_k = \frac{2\pi k}{N}$, $\Delta f = 1/N$
- Aperiodic signals have continuous spectra

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Properties of the DTFT

- Symmetry
- Time-shift
- Time-reversal
- Convolution theorem
- Frequency shifting
- Modulation theorem
- Parseval
- Window theorem

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Properties of the DTFT...

- Symmetry:
- By expressing $x[n]$ in its real and imaginary parts, i.e.,

$$x[n] = x_R[n] + jx_I[n] \xrightarrow{\mathcal{F}} X_R(\omega) + jX_I(\omega)$$
 we can derive a number of symmetry properties
- Example: Real and even signals have real-valued even spectra

$$X(\omega) = X(-\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = X_R(\omega) + j \cdot 0$$
- Check all possibilities: real/imag and even/odd
- Example: $x[n]$ imaginary and odd $\Rightarrow X(\omega)$?

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Properties of the DTFT...

- Answer ($x[n]$ imaginary and odd $\Rightarrow X(\omega)$):

$$x[n] = x_R[n] + jx_I[n] = jx_I[n] \text{ (imaginary)}$$

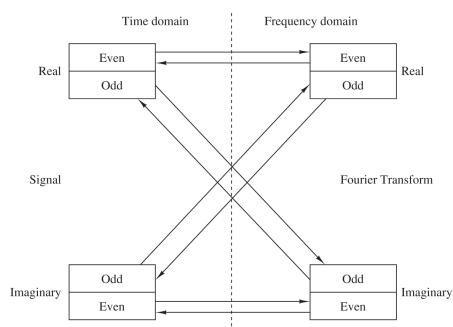
$$x[-n] = -x[n] \text{ (odd)}$$

$$\begin{aligned} X(\omega) &= \sum_{n=-\infty}^{\infty} jx_I[n]e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} jx_I[n](\cos \omega n - j \sin \omega n) \\ &= \sum_{n=-\infty}^{\infty} (jx_I[n] \cos \omega n - j^2 \sin \omega n) \\ &= 2 \sum_{n=0}^{\infty} x_I[n] \sin \omega n \text{ (Real-valued)} \\ \\ X(-\omega) &= 2 \sum_{n=0}^{\infty} x_I[n] \sin[-\omega n] \\ &= -2 \sum_{n=0}^{\infty} x_I[n] \sin[\omega n] = -X(\omega) \text{ (Odd)} \end{aligned}$$

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Properties of the DTFT...



- Rewrite signals in terms of odd and even parts

$$x[n] = (x_R^e[n] + jx_I^e[n]) + (x_R^o[n] + jx_I^o[n])$$

$$X(\omega) = (X_R^e(\omega) + jX_I^e(\omega)) + (X_R^o(\omega) + jX_I^o(\omega))$$

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Properties of the DTFT...

- Time-shift: $x[n - k] \xleftrightarrow{\mathcal{F}} e^{-j\omega k} X(\omega) = |X(\omega)| e^{j(\angle X(\omega) - \omega k)}$
- Time-reversal: $x[-n] \xleftrightarrow{\mathcal{F}} X(-\omega)$
- Convolution: $x_1[n] * x_2[n] \xleftrightarrow{\mathcal{F}} X_1(\omega) X_2(\omega)$
- Frequency shifting: $e^{j\omega_0 n} x[n] \xleftrightarrow{\mathcal{F}} X(\omega - \omega_0)$
- Modulation: $x[n] \cos \omega_0 n \xleftrightarrow{\mathcal{F}} \frac{1}{2} [X(\omega - \omega_0) + X(\omega + \omega_0)]$
- Parseval: $\sum_n |x[n]|^2 \xleftrightarrow{\mathcal{F}} \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(\omega)|^2 d\omega$
- Windowing: $x_1[n] x_2[n] \xleftrightarrow{\mathcal{F}} \frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(\lambda) X_2(\omega - \lambda) d\lambda$

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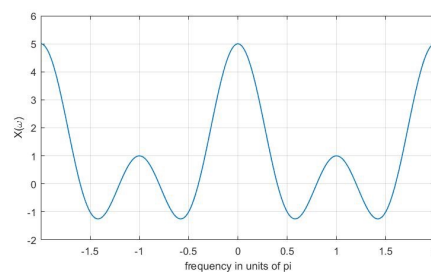
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Properties of the DTFT...

- Example (symmetry): Pulse in time domain

$$x[n] = \{1, 1, \underline{1}, 1, 1\}$$

Sequence $x[n]$ is real and even $\Rightarrow X(\omega)$ is real and even



Matlab

```
n = -2:2; x = ones(1,5);
k = -200:200; w = (pi/100)*k;
X = x * (exp(-j*pi/100)).^(n'*k);
plot(w/pi, real(X)); grid
```

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Properties of the DTFT...

- Example (frequency shift):

$$x[n] = \cos \frac{\pi}{2} n, 0 \leq n \leq 100$$

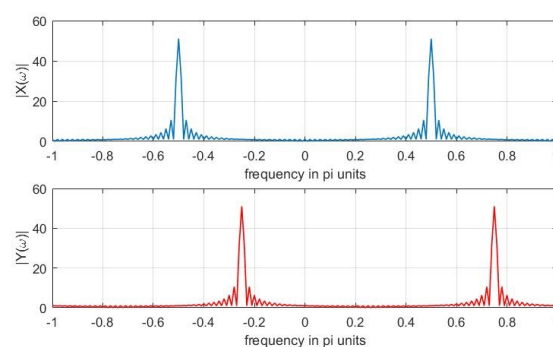
$$y[n] = e^{j\frac{\pi}{4}n} x[n]$$

- Can you guess the shape of the spectra?

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Properties of the DTFT...



Matlab

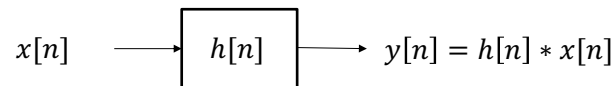
```
n = 0:100; x = cos(pi*n/2);
k = -100:100; w = (pi/100)*k;
X = x * (exp(-j*pi/100)).^(n'*k);
y = exp(j*pi*n/4) .* x;
Y = y * (exp(-j*pi/100)).^(n'*k);
subplot(2,1,1); plot(w/pi,abs(X));
subplot(2,1,2); plot(w/pi,abs(Y));
```

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LTI systems in frequency domain

- Output of linear time-invariant system



- What is the output if the input is a complex exponential?

$$x[n] = Ae^{j\omega n}, \quad -\infty < n < \infty \text{ and } \omega \in [-\pi, \pi]$$

- Compute the convolution

$$\begin{aligned}
 y[n] &= h[n] * x[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] \\
 &= AH(\omega)e^{j\omega n}
 \end{aligned}$$

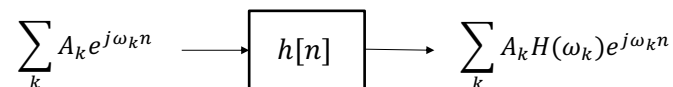
$$\text{with } H(\omega) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} \text{ (frequency response)}$$

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LTI systems in frequency domain...

- Using the linearity of LTI systems



- Frequency response is in general complex-valued

$$H(\omega) = |H(\omega)|e^{j\phi(\omega)}$$

- Magnitude response: $|H(\omega)|$

- Phase response: $\phi(\omega)$

- Group delay: $-\frac{d\phi(\omega)}{d\omega}$

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LTI systems in frequency domain...

- Response to arbitrary input

$$\begin{array}{ccc} x[n] & \longrightarrow & \boxed{h[n]} \longrightarrow y[n] = h[n] * x[n] \\ X(\omega) & & Y(\omega) = H(\omega) X(\omega) \end{array}$$

- Frequency response is in general complex-valued

$$H(\omega) = |H(\omega)|e^{j\phi(\omega)}$$

with magnitude response $|H(\omega)|$ and phase response $\phi(\omega)$

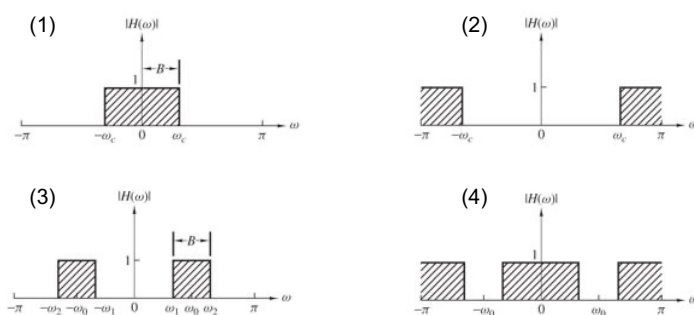
- Frequency response acts like a **spectral shaping** function
- LTI system that performs spectral shaping is referred to as **filter**

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LTI systems in frequency domain...

- Ideal filter characteristics



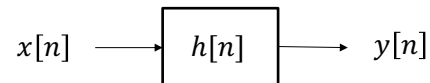
- Practical to implement? What is the time-domain impulse response corresponding to (1)?

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LTI systems in frequency domain...

- Example: LTI system $h[n] = 0.5^n u[n]$ excited by $x[n] = e^{j\frac{\pi}{2}n}$

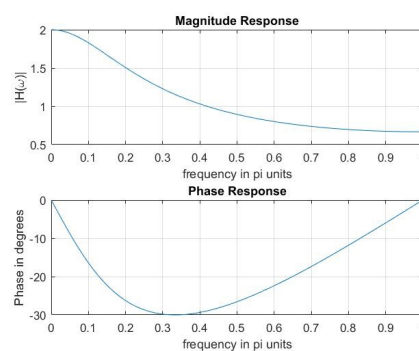


- 1) Compute $y[n]$
- 2) Characterize the type of filter that $h[n]$ represents

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LTI systems in frequency domain...



Matlab
`w = linspace(0, pi, 501);
H = exp(j*w) ./ (exp(j*w) - 0.5);
magH = abs(H); angH = angle(H);
subplot(2,1,1); plot(w/pi,magH); grid;
subplot(2,1,2); plot(w/pi,angH*180/pi); grid`

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Summary

Today:

- Signals and systems in frequency-domain
- Discrete-time Fourier series and transform (DTFS & DTFT)
- Filtering using LTI systems and ideal filters

Next:

- Start the journey of z-transforms

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